

Measuring the Acceleration of the Montreal Metro using an Arduino Train Accelerometer

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Abstract—This study aims to compare acceleration, deceleration, and vibration levels across different metro lines in Montreal (Green, Orange, Yellow, and Blue), different train models (MR-73 and Azur), and individual interstation sections. Using an accelerometer controlled by an Arduino Nano microcomputer, the data collected from 200 interstations revealed significant differences between the train models and metro lines. The older MR-73 trains exhibit higher vibration levels as opposed to the newer Azur trains. For acceleration and deceleration, the Green and Orange lines show similar patterns. The Yellow line exhibits the highest average acceleration and the Blue line the highest deceleration. This research seeks to inform potential improvements in metro systems, contributing to urban transportation studies and ultimately improving passenger experience.

Index Terms—Metro, Public transport, Acceleration analysis, Vibration analysis

I. INTRODUCTION

Most people in urban environments rely heavily on public transportation systems, like the metro, to navigate their cities. Yet, few stop to consider the underlying mechanics of these systems, particularly their acceleration. This paper examines the acceleration and vibration of Montreal metros' from a scientific perspective, using an accelerometer powered by an Arduino microcontroller to collect data. Inspired by daily experiences as Dawson College students commuting on this system, we aim to study the differences between metro lines, train models, and interstation, which are the spaces between two metro stations. The Montreal metro system, operated by the Société de Transport de Montréal (STM) serves as a vital transportation network, consisting of four lines: Green, Orange, Yellow, and Blue. It operates with two primary train models: the MR-73, in service since 1976, and the Azur trains, since 2016 [1]. As of 2025, the MR-73 models operate in the Blue, Green, and Yellow lines, while the Azur models in the Green and Orange lines. In this study, acceleration will be measured along three axes: X, Y, and Z. These measurements are influenced by two types of forces: inertial acceleration,

caused by changes in velocity, and gravitational acceleration, resulting from the Earth's gravitational field. For instance, when an accelerometer is placed in the metro, it measures the gravitational acceleration, g , in addition to the inertial acceleration caused by the train's motion. Assuming a flat surface, gravitational acceleration can be subtracted from the total measurement to isolate inertial acceleration. However, the problem becomes more complex on an inclined plane. Here, the gravitational component along the incline is $g \sin(\theta)$, where $g = 9.81 \text{ m/s}^2$ is Earth's gravitational constant and θ is the angle of inclination. As per Newton's equivalence principle, without additional data, such as θ , it becomes impossible to determine how much of the vertical acceleration is due to gravity versus the metro's inertial motion i.e. if the acceleration measured is due to the metro's motion or to the changing slope [2, 3]. Concerning vibrations and lateral curvature, these metrics provide insights into the metro's operation and passenger experience. Vibrations, quantified in this paper as the standard deviation of acceleration, allow for the comparison of the levels of comfort across different metro lines and train models. For example, newer Azur trains are expected to produce smoother rides than the older MR-73 models. Furthermore, lateral acceleration measurements on the X and Y axes can reveal track curvature. Assuming the metro train remains upright, these measurements indicate whether the train is turning left or right. By identifying patterns in vibrations and curvature, we can assess the impact of track design and train model on ride quality. By collecting these metrics, the acceleration and vibration levels of the Montreal metro system can be analyzed. Specifically, this study aims to compare acceleration, deceleration, and vibration levels across different metro lines (Green, Orange, Yellow, and Blue), different train models (MR-73 and Azur), and individual interstation sections. This way, this research seeks to inform potential improvements in metro systems, contributing to urban transportation studies and ultimately improving passenger experience.

II. METHODS

A systematic approach was implemented to collect, calibrate, and analyse acceleration and vibration data of Montreal's metro system.

A. Device

The device needs to be portable as the data must be collected on board the metro. It consists of a compact electronic circuit inside a cardboard box, small enough to fit under a metro seat. During data collection, the box must remain stationary, otherwise the acceleration data is not reliable. Some strips of anti-slip tape on the bottom of the box ensure a high enough friction between the box and the floor to stop it from moving.

B. Circuit

The circuit, shown in fig. 1, is composed of an Arduino Nano microcontroller, an MPU-6050 three-axis accelerometer, a light-emitting diode (LED), a switch, and a memory card (micro-SD card), all powered by a 9-volt battery. The battery is only connected to the microcontroller before a period of data collection, called a run, to save battery power. The connections are made using a breadboard. The breadboard and microcontroller are both glued to the box to prevent unwanted motion and acceleration in the data. When power is received from the 9V battery, if a component is not correctly wired, the LED flashes red, otherwise it lights up blue. After this point, flipping the switch starts the data collection. The accelerometer is oriented perpendicular to the breadboard and parallel to one side of the box, as shown in fig. 2, on the right of the breadboard. This ensures that the three axes of the accelerometer correspond to the three physical axes of the box. These are defined in the following way: the X axis is vertical, the Y axis is longitudinal (forwards and backwards), and the Z axis is lateral (right and left). As the accelerometer is firmly kept in place by its multiple connections to the breadboard, its orientation relative to the box is assumed to be constant. One side of the box is designated as the front, such that accelerating in that direction registers a positive Y acceleration.

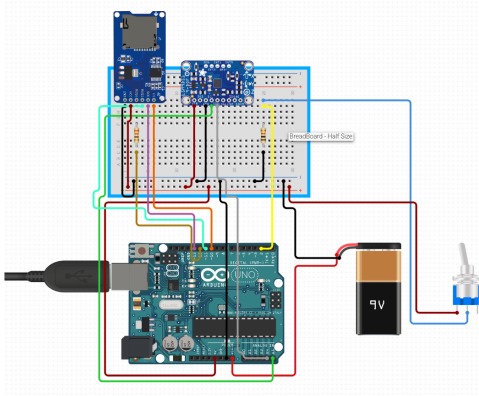


Fig. 1. Circuit wiring diagram made with Circuito®

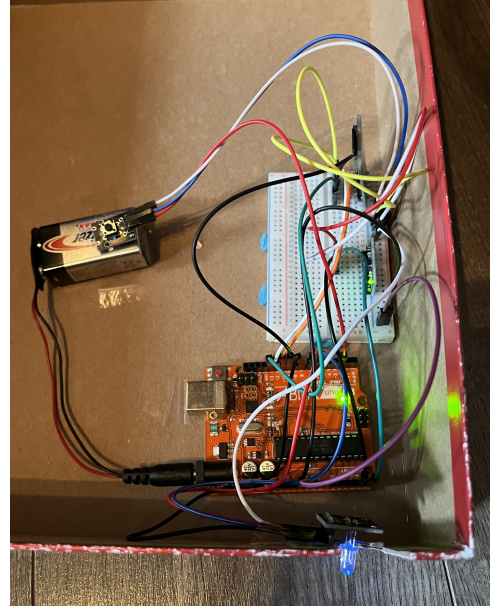


Fig. 2. Fully wired circuit in the open box. The LED, visible at the bottom right of the image, is blue, indicating the circuit is ready for data collection.

C. Procedure

Every continuous period during which data is collected on the metro is called a run. Before each run, the circuit is powered by connecting the 9V battery. Once the LED indicates normal functioning, the box is placed on the floor of the metro, usually under a seat. The front of the box is always located in the direction of forward motion, with one side of the box against the wall to ensure the device is parallel to the metro. The switch is then flipped to start data collection. A data point consisting of the acceleration value for each of the X, Y, and Z directions is then recorded by the device 10 times per second. During a run, the experimenter records information about the run to categorize it, including the date, type of train, metro line, and start and end stations, as well as information that can later be used to compare the context of different runs, such as number of people in the train, position in the train and in the train car. A run can last until the metro reaches the end of a line, or at any station before that point. To stop a run, the box is picked up and the switch is flipped. It is necessary to wait at least two seconds after the metro comes to rest before interacting with the device, since the acceleration is registered with some delay. Each run is saved in a separate comma-separated values (CSV) file on the micro-SD card. In total, 20 runs were done in the four lines of the metro, totaling 200 interstations. 95 of these interstations were on the green line, 74 on the orange line, 10 on the yellow line, and 21 on the blue line. 128 of the interstations were in Azur trains and 72 in MR-73 trains.

1) *Data analysis:* Each CSV file is converted to a spreadsheet in Excel®. All data points before the first significant acceleration period of a run and after the last significant deceleration are deleted so that the run begins and ends with

only a short period at rest. This procedure is necessary for proper calibration of the acceleration.

2) *Calibration:* Data collected from the accelerometer has an offset at rest from the expected 0 m/s². The average value of a function over an interval is equal to its antiderivative divided by the length of the interval:

$$a_{x,avg} = \frac{1}{L} \int_0^L a_x dx \quad (1)$$

with a_x being the acceleration in any of the three axes. The antiderivative of inertial acceleration is velocity, so the average value over the entire run is equal to the change in speed over the run, divided by the time of the run. Since it is specified that the run begins and ends at rest, the average value is zero. However, the accelerometer records inertial and gravitational acceleration together, so this formula is not exact. To simplify the calculations, gravitational acceleration is assumed to be 0. This only slightly affects the Y and Z acceleration as the angle between the train and the horizontal is not significant ($<3.76^\circ$). With this assumption, each axis of each run is calibrated as detailed.

3) *Acceleration and deceleration:* To reduce the noise in the data, the average longitudinal acceleration over 10 data points is taken for the entire run. For each interstation, the highest and lowest values are recorded. An example of longitudinal acceleration for an interstation is provided in fig. 3.

4) *Vibration:* A method is devised to quantify the vibration level of each interstation from the acceleration data. Vibration is estimated using the standard deviation of the lateral and vertical accelerations because as the train vibrates, the accelerometer will record rapidly changing values. This method is arbitrary, as different parameters could lead to different values, but comparing all the data using the same method still allows for meaningful comparison. The standard deviation of both accelerations is taken separately for a moving interval of 100 points, then the resulting data series are smoothed by taking a moving average of 255 points as shown by fig. 4. The highest value each series reaches during each interstation is defined as the vibration level, on the lateral or vertical axis, of that interstation for the run and recorded as one data point.

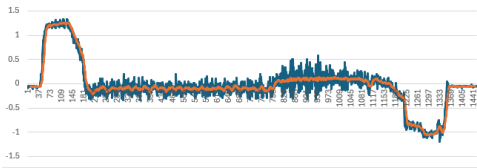


Fig. 3. Graph of longitudinal acceleration for one interstation, with the calibrated data in blue and its average over 10 data points in orange. The highest and lowest points of the orange curve over this interval are the maximum and minimum acceleration values for that interstation. The maximum occurs upon initial acceleration and the minimum during the final deceleration of the interstation.

5) *Lateral curvature:* An accurate map of the Montreal metro network is used to qualitatively match the maxima and minima of the lateral acceleration to physical characteristics of

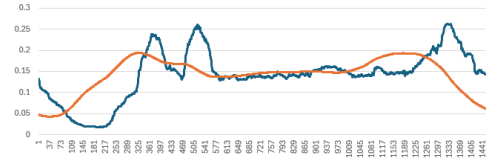


Fig. 4. Graph of lateral vibration for the same interstation, with the 100 data points standard deviation in blue and its 255-point average in orange. The highest point of the orange curve over this interval is the lateral vibration level for that interstation. The two curves are offset due to the difference in number of data points being averaged.

the path. Positive values are associated with curvature to the left and negative values to the right. No exhaustive analysis is done for this component of research, since its goal is only to determine whether the device is a viable way to obtain data for that purpose.

III. RESULTS

A comparative analysis of acceleration and vibration levels across different metro lines and models is presented below.

A. Acceleration and deceleration

The highest average acceleration is observed on the Yellow line (1.30 ± 0.04 m/s²), while the Blue line has the lowest (1.24 ± 0.05 m/s²). The Green (1.28 ± 0.06 m/s²) and Orange (1.27 ± 0.03 m/s²) lines show nearly identical acceleration values. In terms of deceleration, the Blue line registers the highest average (-1.23 ± 0.15 m/s²), whereas the Yellow line has the lowest (-1.07 ± 0.24 m/s²). A comparison between train models shows that the Azur and MR-73 trains have similar acceleration values (1.27 ± 0.04 m/s² and 1.28 ± 0.05 m/s², respectively). However, the Azur reveals a lower variability, and thus a more consistent performance. In contrast, MR-73 trains exhibit a higher average deceleration (-1.25 ± 0.19 m/s²) compared to Azur (-1.14 ± 0.22 m/s²). Measured acceleration values for Azur trains frequently exceed the expected theoretical maximum (1.207 m/s²) with some data points also surpassing the expected deceleration limit (-1.790 m/s²) [4]. In contrast, MR-73 acceleration and deceleration values align more closely with theoretical ones (1.431 m/s² and -1.788 m/s², respectively) [5]. A complete overview of the data is provided in fig. 5 and in fig. 6.

B. Vibration

Vibration levels vary significantly between train models and metro lines. The MR-73 trains, as well as the lines primarily operating them, exhibit higher vibration levels as opposed to Azur trains. The Yellow line has the highest vibration levels, while the Orange line has the lowest as shown in fig. 7 and in fig. 8.

C. Lateral curvature

Lateral acceleration patterns correspond qualitatively with the curvature of the metro tracks. These measurements are due to the radial acceleration necessary in any curve and to the fact that tracks are often tilted towards the center of curvature.

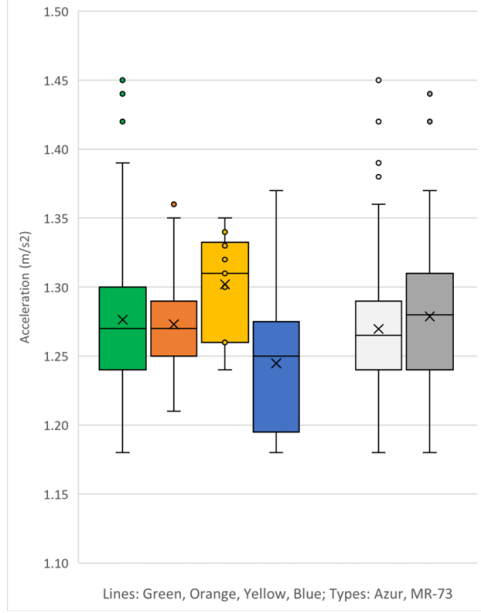


Fig. 5. Box plot (whiskers show 1.5 IQR) for the maximum acceleration for each interstation of the metro. The median is indicated by a horizontal line, the average by a cross. Outliers are shown for all categories, all points for the Yellow line are shown because of the small sample size ($n=10$)

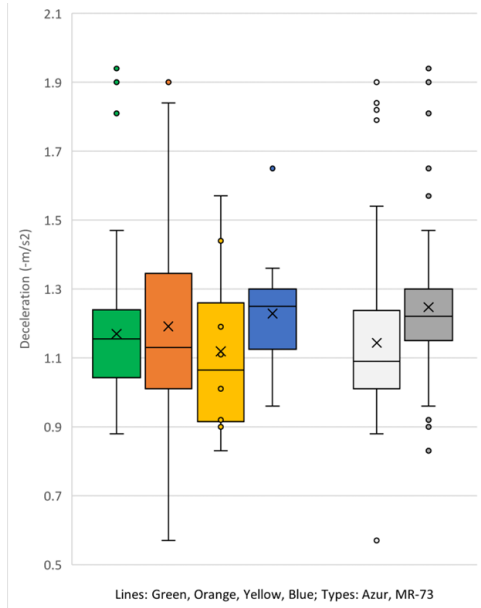


Fig. 6. Box plots for the maximum deceleration for each interstation of the metro. Parameters are the same as for fig. 5

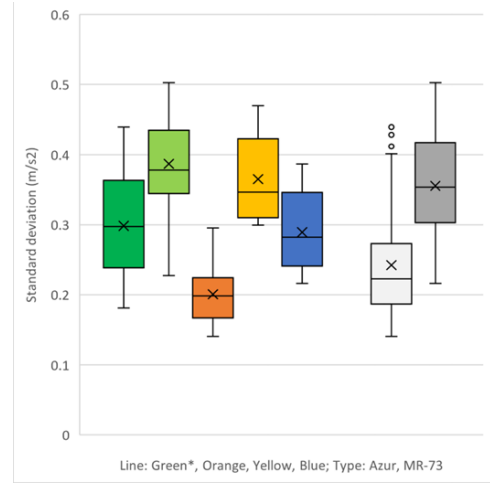


Fig. 7. Box plot for lateral vibration level for each interstation. The data from the Green line are separated between the Azur (left) and MR-73 (right) models due to a significant difference. Other parameters are the same as for fig. 5.

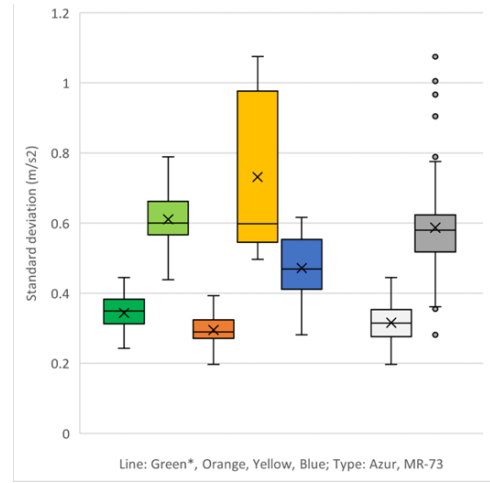


Fig. 8. Box plot for vertical vibration level for each interstation. Parameters are the same as for fig. 7.

Both phenomena are sensed by the accelerometer in the lateral axis. By comparing acceleration peaks to a reference map of the Montreal metro system shown in fig. 9, turns and straight segments can be inferred from the recorded data [6].

IV. DISCUSSION

The main goal of this research was to discover whether the Montreal metro models and lines significantly vary in a measurable way. To do so, various runs were done, as described in Methods, and the data collected were compared. Quantifying acceleration and vibration levels confirmed the presence of these differences. The findings generally agree with the theoretical data and information obtained from the STM. Multiple methods were employed to ensure data analysis, one of which is data calibration. This process was necessary due to Earth's gravitational field, which offsets the Y axis's data by g . Vertical acceleration was thus calibrated to, on

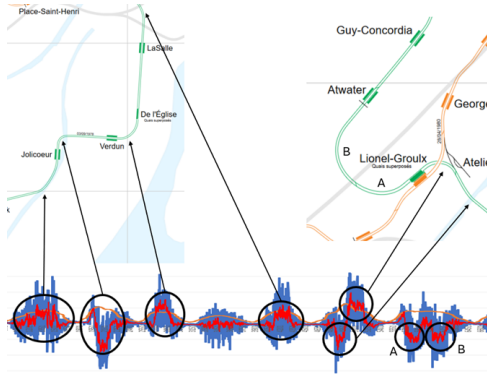


Fig. 9. Sections of Montreal's metro system, represented in a geographical map, compared to the lateral acceleration measurements.

average, lie at 0, such that any upwards or downwards motion performed by the metro would be measured separately from the gravitational acceleration. This does not affect the results obtained, while enabling the data to reflect more accurately how the train behaves, allowing thus for the graphs to represent the situation in a more visual manner. Much of the data was collected from one non-terminus station to another non-terminus station. This meant that data from different stations belonging to the same line can be found across multiple different files, which may have been measured in slightly different conditions. To take this into account, a way to atomise the data to be collected was adopted. By measuring and calculating acceleration and vibration for each interstation rather than averaging over entire metro lines, information was extracted from small units. In summary, the calibration of vertical acceleration provided more accurate data, allowing for clearer visualization of the train's behavior. Furthermore, segmenting data at the interstation level enhanced the precision of the findings, offering valuable insights into the metro's motion.

V. CONCLUSION

The data shows that there are quantifiable variations between the different trains and lines. For maximum acceleration, the Yellow line exhibits the highest values, while the Blue line, the lowest. As for maximum deceleration, the inverse occurs. In both cases, the Green and Orange lines are approximately equal to each other. Concerning vibrations, the MR-73 has distinctly higher levels as compared to the Azur train. The results also show the vertical vibration levels being significantly higher than the lateral ones. Observations of the lateral acceleration data also made it possible to detect curves in the metro's path. Finally, the experimental values relating to acceleration mostly exceed the theoretical values.

A. Limitations

The metro does not travel on flat ground in the tunnels, thus a component of gravitational acceleration is registered as inertial acceleration. This can explain values higher than their theoretical maximum. The dataset obtained is potentially

not representative of the entire metro network. The number of downtown interstations and runs during rush hour is higher, for example, due to our process of collection.

B. Further research

Using an Arduino-compatible gyroscope would enable the determination of the angular orientation of the metro at any time, thus allowing for a further analysis of acceleration. Once this is accomplished, a study of the REM could also be conducted, since its tracks often have a non-negligible deviation from the horizontal plane, particularly in curves. More advanced mathematical tools, like determination of the radius of curvature of a curve, could then be used to gather more quantitative data about the physical layout of the tracks, rather than qualitative as is currently the case. Numerical integration of the longitudinal acceleration could make calculating the speed reached by the metro possible, giving more data to compare to theoretical values. However, this is only possible once the discrepancy between inertial and gravitational acceleration is taken into account. With more data, it would be possible to compare the acceleration and vibration differences between older and newer tracks, since this information is available. This could lead to an explanation for the difference in vibration between different interstations.

ACKNOWLEDGEMENT

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